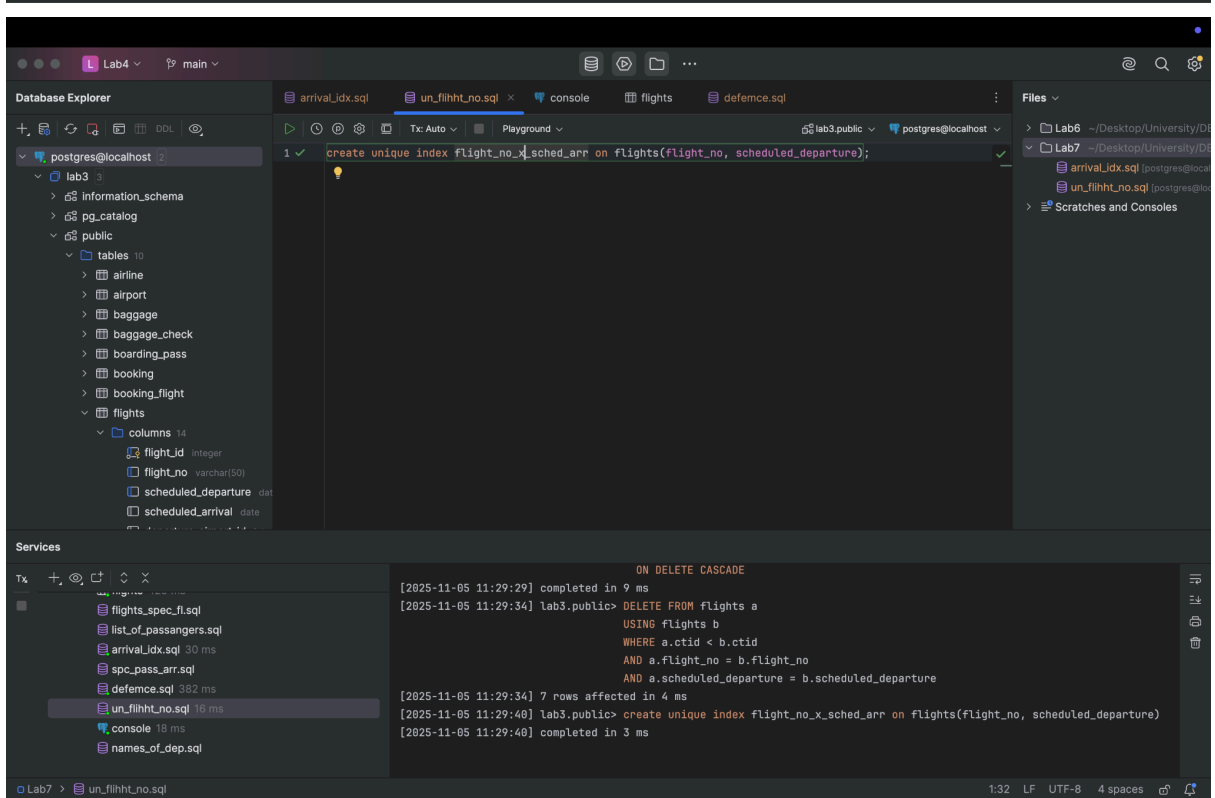




Lab4 main

Database Explorer

- postgres@localhost
 - lab3
 - information_schema
 - pg_catalog
 - public
 - tables 10
 - airline
 - airport
 - baggage
 - baggage_check
 - boarding_pass
 - booking
 - booking_flight
 - flights
 - columns 14
 - flight_id integer
 - flight_no varchar(50)
 - scheduled_departure date
 - scheduled_arrival date

Files

- Lab6 ~/Desktop/University/Dt
- Lab7 ~/Desktop/University/Dt
 - arr_dep_air.sql [postgres@loc
 - arrival_idx.sql [postgres@loc
 - un_flihht_no.sql [postgres@loc
 - Scratches and Consoles

Services

1 create index dep_arr_airport on flights(departure_airport_id, arrival_airport_id);

2025-11-05 11:30:55] Connected to lab3
[2025-11-05 11:30:55] completed in 26 ms
[2025-11-05 11:31:25] lab3> create index dep_arr_airport on flights(departure_airport_id, arrival_airport_id)
[2025-11-05 11:31:25] completed in 6 ms

list_of_passangers.sql
arrival_idx.sql 30 ms
spc_pass_arr.sql
defemce.sql 382 ms
un_flihht_no.sql 16 ms
console 18 ms
arr_dep_air.sql 25 ms
names_of_dep.sql

Lab7 > arr_dep_air.sql 1:82 LF UTF-8 4 spaces

Lab4 main

Database Explorer

- postgres@localhost
 - lab3
 - information_schema
 - pg_catalog
 - public
 - tables 10
 - airline
 - airport
 - baggage
 - baggage_check
 - boarding_pass
 - booking
 - booking_flight
 - flights
 - columns 14
 - flight_id integer
 - flight_no varchar(50)
 - scheduled_departure date
 - scheduled_arrival date

Files

- Lab6 ~/Desktop/University/Dt
- Lab7 ~/Desktop/University/Dt
 - arr_dep_air.sql [postgres@loc
 - arrival_idx.sql [postgres@loc
 - un_flihht_no.sql [postgres@loc
 - Scratches and Consoles

Services

1 create index dep_arr_airport on flights(departure_airport_id, arrival_airport_id);
2
3 drop index dep_arr_airport;
4
5 \timing on
6 select flights.flight_no from flights;

Output lab3.public.flights lab3.public.flights 2

[2025-11-05 11:36:17] lab3.public> select flights.flight_no from flights
[2025-11-05 11:36:17] 500 rows retrieved starting from 1 in 350 ms (execution: 3 ms, fetching: 347 ms)
[2025-11-05 11:36:35] lab3.public> select flights.flight_no from flights
[2025-11-05 11:36:35] 500 rows retrieved starting from 1 in 331 ms (execution: 4 ms, fetching: 327 ms)
[2025-11-05 11:37:01] lab3.public> create index dep_arr_airport on flights(departure_airport_id, arrival_airport_id)
[2025-11-05 11:37:01] completed in 3 ms
[2025-11-05 11:37:05] lab3.public> select flights.flight_no from flights
[2025-11-05 11:37:06] 500 rows retrieved starting from 1 in 329 ms (execution: 6 ms, fetching: 323 ms)

list_of_passangers.sql
arrival_idx.sql 30 ms
spc_pass_arr.sql
defemce.sql 382 ms
un_flihht_no.sql 16 ms
console
arr_dep_air.sql 342 ms
names_of_dep.sql

Lab7 > arr_dep_air.sql 6:39 LF UTF-8 4 spaces

Lab4main

Database Explorer

postgres@localhost2

lab33

information_schema

pg_catalog

public

tables10

airline

airport

baggage

baggage_check

boarding_pass

booking

booking_flight

flights

columns14

flight_idinteger

flight_novarchar(50)

scheduled_departuredate

scheduled_arrivaldate

arrival_idx.sql

un_flihht_no.sql

arr_dep_air.sql

exp_analyze.sql

defence.sql

console

1

2

3

4

5

6

create index depp_arr_airport

on flights(departure_airport_id, arrival_airport_id);

explain analyse

select flights.flight_no, flights.flight_id from flights

where departure_airport_id = '2' and arrival_airport_id = '4';

Files

Lab6~/Desktop/University/Dt

Lab7~/Desktop/University/Dt

arr_dep_air.sql[postgres@loc

arrival_idx.sql[postgres@local

defence.sql[postgres@localh

exp_analyze.sql[postgres@lo

un_flihht_no.sql[postgres@loc

Scratches and Consoles

Services

1%

Database

postgres@localhost

arr_dep_air.sql399 ms

arrival_idx.sql

un_flihht_no.sql

exp_analyze.sql337 ms

defence.sql

console

Output

Result 4

CSV

7 rows

QUERY PLAN

1 Bitmap Heap Scan on flights (cost=4.31..12.83 rows=3 width=10) (actual time=0.014..0.015 rows=1 loops=1)

2 Recheck Cond: ((departure_airport_id = 2) AND (arrival_airport_id = 4))

3 Heap Blocks: exact=1

4 -> Bitmap Index Scan on depp_arr_airport (cost=0.00..4.31 rows=3 width=0) (actual time=0.009..0.009 rows=1 loops=1)

5 Index Cond: ((departure_airport_id = 2) AND (arrival_airport_id = 4))

6 Planning Time: 0.063 ms

Lab7exp_analyze.sql6:63 LF UTF-8 4 spaces

The screenshot shows a database management tool interface. On the left, the 'Database Explorer' pane shows a PostgreSQL database with a 'public' schema containing a 'passengers' table. The 'passengers' table has columns: flight_id (integer), flight_no (varchar(50)), scheduled_departure (date), and scheduled_arrival (date). The main editor shows the following SQL code:

```
1 create index pass_num
2 on passengers(passport_number);
3
4 select indexname, indexdef
5 from pg_indexes
6 where tablename = 'passengers';
7
8
```

The 'Output' pane at the bottom right shows the result of the query:

indexname	indexdef
passengers_pkey	CREATE UNIQUE INDEX passengers_pkey ON public.passengers USING btree (passenger_id)
un_passport	CREATE UNIQUE INDEX un_passport ON public.passengers USING btree (passport_number)
pass_num	CREATE INDEX pass_num ON public.passengers USING btree (passport_number)

The screenshot shows the same database management tool interface. The main editor now shows the following SQL code:

```
4 on passengers(passport_number);
5
6 select indexname, indexdef
7 from pg_indexes
8 where tablename = 'passengers';
9
10 insert into passengers values
11 (passenger_id '10001', first_name 'Nikita', last_name 'Lee', date_of_birth '2006-01-01');
12
13 insert into passengers values
14 (passenger_id '10002', first_name 'Nikita', last_name 'Lee', date_of_birth '2006-01-01');
15
16 delete
17 from passengers
18 where passenger_id = '10001' or passenger_id = '10002';
19
```

The 'Output' pane at the bottom right shows the result of the query:

indexname	indexdef
passengers_pkey	CREATE UNIQUE INDEX passengers_pkey ON public.passengers USING btree (passenger_id)
un_passport	CREATE UNIQUE INDEX un_passport ON public.passengers USING btree (passport_number)
pass_num	CREATE INDEX pass_num ON public.passengers USING btree (passport_number)

A red error message is displayed at the bottom of the editor:

```
[23505] ERROR: duplicate key value violates unique constraint "pass_num"
Detail: Key (passport_number)=(KZ123) already exists.
```

It doesn't give opportunity to insert 2 passengers with the same id because passport number must be unique and I've created a unique index for this one. So, it throws an error to avoid mistakes in table and index could work correctly, because each cell of passport_number must be unique.

Database Explorer

```

1 create index adv_idx
2 on passengers(first_name, last_name, date_of_birth, country_of_citizenship);
3
4
5 explain analyse
6 select passengers.passenger_id, passengers.first_name || ' ' || passengers.last_name as fullname, country_o
7 where country_of_citizenship = 'Philippines' and
8 date_of_birth = date '1984-01-01';
9
10

```

Services

Output

```

1 Seq Scan on passengers (cost=0.00..6.02 rows=1 width=44) (actual time=0.031..0.031 rows=0 loops=1)
2   Filter: (((country_of_citizenship)::text = 'Philippines'::text) AND (date_of_birth = '1984-01-01'::date))
3   Rows Removed by Filter: 201
4 Planning Time: 0.076 ms
5 Execution Time: 0.043 ms

```

it didnt use index because table is not big enough.

Database Explorer

```

1 select indexname, pg_indexes.indexdef from pg_indexes
2 where tablename = 'passengers';
3
4
5 drop index adv_idx;
6 drop index pass_num;
7

```

Services

Output

```

1 indexname      : indexdef
2 passengers_pkey CREATE UNIQUE INDEX passengers_pkey ON publ_
3 pass_num       CREATE UNIQUE INDEX pass_num ON public.pass_
4 adv_idx        CREATE INDEX adv_idx ON public.passengers U_

```